Information-Theoretic Observation Geometry

Entropy Defect, Fiber Uncertainty, Data Processing, and Recoverable Invariants

Stassis Research Program

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Abstract

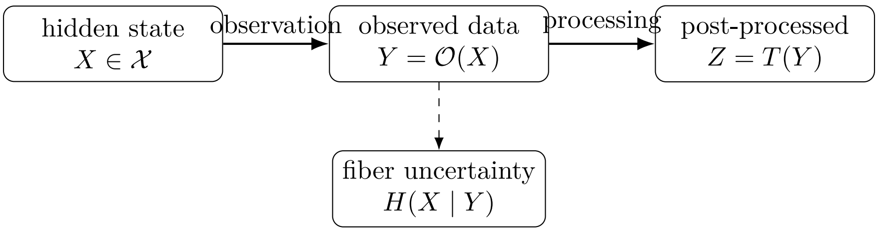
This note introduces an information-theoretic layer for Geometry of Observation. The central object is an observation map or channel from hidden states to observed data. Shannon entropy measures finite distinguishability, conditional entropy measures hidden uncertainty remaining inside observation fibers, mutual information measures recoverable statistical information, and data processing formalizes the fact that post-processing cannot increase distinguishability. We define an observational entropy defect and connect it to recoverable invariants, half-step aliasing, hidden compact fibers, and metric entropy. The note is deliberately narrow: it does not claim a physical theory of mass, gravity, or consciousness. It supplies a mathematical vocabulary for the statement that observation loses not merely coordinates, but distinguishability.

Contents

1 Scope and claim firewall

The purpose of this note is to make the probabilistic information layer of Geometry of Observation explicit. Earlier notes introduced observation maps, entropy defects based on covering numbers, matroidal rank defects, and constraint-satisfaction frustration. Here the same idea is expressed in the standard language of information theory.

Claim firewall. The note does not claim that Shannon entropy is physical entropy in every context, that information theory derives general relativity, or that an observation map creates reality. The admissible claim is narrower: if hidden states are modeled probabilistically, then observation maps and channels have precise information losses measured by conditional entropy, mutual information, and divergence contraction.



Post-processing cannot increase information about the hidden source; it can only retain, discard, or re-encode it.

Figure 1: Observation as an information channel. The hidden state X is not identified with the observed datum Y. Conditional entropy measures remaining hidden distinguishability inside observation fibers.

2 Observation channels

Definition 2.1 (Finite hidden and observed spaces)**.** For the finite theorem layer, let $\mathcal{X}$ be a finite hidden state space and let $\mathcal{Y}$ be a finite observation space. A hidden distribution is a probability mass function P_X on $\mathcal{X}$.

Definition 2.2 (Deterministic observation map)**.** A deterministic observation is a map

$$\mathcal{O} : \mathcal{X} \to \mathcal{Y}.$$

It induces the observed random variable

$$Y = \mathcal{O}(X).$$

The fiber over $y \in \mathcal{Y}$ is

$$\mathcal{O}^{-1}(y) = \{x \in \mathcal{X} : \mathcal{O}(x) = y\}.$$

Definition 2.3 (Stochastic observation channel)**.** A stochastic observation channel is a Markov kernel

$$K(y \mid x), \qquad x \in \mathcal{X}, y \in \mathcal{Y},$$

with $K(y \mid x) \geq 0$ and $\sum_y K(y \mid x) = 1$. It induces the joint law

$$P_{X,Y}(x, y) = P_X(x)K(y \mid x).$$

A deterministic observation is the special case $K(y \mid x) = \mathbf{1}_{\{y=\mathcal{O}(x)\}}$.

Remark 2.4 (Observation versus coordinate change). A coordinate change is typically invertible on its domain. An observation channel need not be invertible. It can merge hidden states, add noise, erase features, or coarsen a continuum of states into a finite alphabet.

3 Entropy, mutual information, and fiber uncertainty

All logarithms are natural unless another base is explicitly stated.

Definition 3.1 (Shannon entropy)**.** For a finite random variable X, define

$$H(X) = -\sum_{x\in\mathcal{X}} P_X(x)\log P_X(x),$$

with the convention $0\log 0 = 0$.

Definition 3.2 (Conditional entropy and mutual information)**.** For jointly distributed finite random variables (X, Y), define

$$H(X \mid Y) = \sum_{y\in\mathcal{Y}} P_Y(y) H(X \mid Y = y)$$

and

$$I(X; Y) = H(X) - H(X \mid Y) = H(Y) - H(Y \mid X).$$

Theorem 3.3 (Deterministic observation entropy decomposition)**.** *If $Y = \mathcal{O}(X)$ is deterministic, then*

$$H(X) = H(Y) + H(X \mid Y).$$

Consequently,

$$I(X; Y) = H(Y), \qquad H(Y) \le H(X).$$

Proof. Since Y is a deterministic function of X, $H(Y \mid X) = 0$. The chain rule gives

$$H(X, Y) = H(X) + H(Y \mid X) = H(X),$$

and also

$$H(X, Y) = H(Y) + H(X \mid Y).$$

Equating the two expressions gives the first identity. The statements for mutual information follow from the definition. □

Definition 3.4 (Observational entropy defect)**.** For a deterministic observation $\mathcal{O} : X \mapsto Y$, define

$$\boxed{\mathfrak{D}_{\mathcal{O}}(X) := H(X \mid Y).}$$

This is the expected hidden distinguishability remaining inside the fibers of $\mathcal{O}$.

Proposition 3.5 (Fiber formula)**.** *For deterministic $Y = \mathcal{O}(X)$,*

$$\mathfrak{D}_{\mathcal{O}}(X) = \sum_{y: P_Y(y)>0} P_Y(y) \left[-\sum_{x\in\mathcal{O}^{-1}(y)} P(x \mid y)\log P(x \mid y) \right].$$

Thus the entropy defect is the average entropy of the observation fibers.

Corollary 3.6 (Zero defect criterion)**.** *For deterministic observation, $\mathfrak{D}_{\mathcal{O}}(X) = 0$ if and only if X is determined by Y on the support of P_X. Equivalently, every observed fiber intersecting the support contains exactly one hidden state with positive conditional probability.*

4 Data processing

The defining law of information-theoretic observation is that post-processing cannot create information about the hidden source.

Theorem 4.1 (Data processing inequality)**.** *If*

$$X \to Y \to Z$$

is a Markov chain, then

$$I(X;Z) \leq I(X;Y).$$

Proof. Using the chain rule for mutual information,

$$I(X;Y,Z) = I(X;Z) + I(X;Y \mid Z)$$

and also

$$I(X;Y,Z) = I(X;Y) + I(X;Z \mid Y).$$

The Markov condition $X \to Y \to Z$ gives $I(X;Z \mid Y) = 0$. Therefore

$$I(X;Z) + I(X;Y \mid Z) = I(X;Y).$$

Since conditional mutual information is nonnegative, $I(X;Z) \leq I(X;Y)$. □

Definition 4.2 (Information loss under a channel)**.** For a channel $K : \mathcal{X} \to \mathcal{Y}$, define the retained information as $I(X;Y)$ and the unrecovered hidden uncertainty as $H(X \mid Y)$. If the channel is deterministic, this reduces to the fiber entropy defect of the previous section.

5 Divergences and distinguishability cost

Entropy measures uncertainty of one distribution. Divergences compare two distributions.

Definition 5.1 (Kullback–Leibler divergence)**.** For probability distributions P, Q on a finite set $\mathcal{X}$, define

$$\mathrm{D}_{\mathrm{KL}}(P\|Q) = \sum_{x\in\mathcal{X}} P(x)\log\frac{P(x)}{Q(x)},$$

when P is absolutely continuous with respect to Q.

Theorem 5.2 (KL contraction under observation)**.** *Let K be a stochastic observation channel. Let KP and KQ be the observed distributions on $\mathcal{Y}$. Then*

$$\mathrm{D}_{\mathrm{KL}}(KP\|KQ) \leq \mathrm{D}_{\mathrm{KL}}(P\|Q).$$

Proof. This is the log-sum inequality. For each y set $a_x = P(x)K(y \mid x)$ and $b_x = Q(x)K(y \mid x)$. Then

$$\sum_x a_x \log\frac{a_x}{b_x} \geq \left(\sum_x a_x\right)\log\frac{\sum_x a_x}{\sum_x b_x}.$$

Summing over y gives the result, because the left side becomes $\mathrm{D}_{\mathrm{KL}}(P\|Q)$ and the right side becomes $\mathrm{D}_{\mathrm{KL}}(KP\|KQ)$. □

Remark 5.3 (Meaning for observation geometry)*.* Two hidden distributions may be easier to distinguish before observation than after observation. A channel can preserve, reduce, or erase distinguishability, but it cannot increase KL distinguishability.

6 Recoverable invariants

Information theory gives a probabilistic version of the fiber-factorization criterion used in Geometry of Observation.

Definition 6.1 (Recoverable invariant)**.** Let $Q = f(X)$ be a finite-valued hidden invariant. It is recoverable from Y if there exists a function $\widehat{f}$ such that

$$Q = \widehat{f}(Y)$$

almost surely.

Theorem 6.2 (Entropy criterion for recoverability)**.** *A finite-valued invariant $Q = f(X)$ is recoverable from Y if and only if*

$$H(Q \mid Y) = 0.$$

If $Y = \mathcal{O}(X)$ is deterministic, this is equivalent to f being constant on every observation fiber intersecting $\operatorname{supp} P_X$*.*

Proof. For finite variables, $H(Q \mid Y) = 0$ if and only if the conditional law of Q given $Y = y$ is a point mass for every y with positive probability. Hence there is a measurable function $\widehat{f}$ with $Q = \widehat{f}(Y)$ almost surely. In the deterministic case, the conditional law over a fiber is a point mass precisely when f is constant on that fiber, up to null states. □

Corollary 6.3 (Observation can preserve an invariant while losing the state)**.** *It is possible that $H(X \mid Y) > 0$ but $H(f(X) \mid Y) = 0$. Thus full hidden-state recovery may fail while a declared invariant remains exactly recoverable.*

7 Examples

7.1 Half-step aliasing

Let the hidden index be uniform on

$$\mathcal{X}_m = \{0, 1, \dots, 2m - 1\}$$

and let the observation record only parity:

$$Y = n \bmod 2.$$

This is the finite-window version of the half-step trace

$$d_n = d_0 + \frac{n}{2}, \qquad h(n) = (-1)^n.$$

Then

$$H(X) = \log(2m), \qquad H(Y) = \log 2,$$

and therefore

$$\boxed{H(X \mid Y) = \log m.}$$

The observed trace retains one bit of parity and loses the remaining hidden index within each parity class.

7.2 Hidden compact fiber

Let

$$X = (B, F), \qquad Y = B,$$

with B and F independent finite variables. Then

$$H(X \mid Y) = H(F).$$

If F is an m-bin discretization of a hidden circle S^1, then

$$H(F) = \log m.$$

As the angular resolution is refined, $m \sim 2\pi/\varepsilon$ and

$$H(F) \sim \log \frac{1}{\varepsilon} + \log(2\pi).$$

This is the information-theoretic analogue of the metric entropy loss of a hidden compact S^1 fiber.

7.3 Chromatic cell field

Let $P = \{p_1, \dots, p_n\}$ be metric cells and let

$$c : P \to \mathcal{C}$$

be a color, phase, or code assignment. If the hidden state is $(p, c(p))$ but the observation records only p, then

$$H(p, c(p) \mid p) = H(c(p) \mid p).$$

If the color is determined by position, the defect is zero. If several hidden colors are compatible with one observed position, the defect measures unresolved chromatic multiplicity.

8 Continuous variables and metric entropy

Remark 8.1 (Differential entropy caveat). For continuous variables, differential entropy

$$h(X) = -\int p(x) \log p(x) \, \mathrm{d}x$$

is not invariant under coordinate changes and can be negative. In observation geometry, mutual information, KL divergence, discretized entropy at resolution ε, and metric entropy are usually safer objects.

Let $N_\varepsilon(X, d)$ be the covering number of a compact metric space and

$$H_\varepsilon(X, d) = \log N_\varepsilon(X, d).$$

The finite probabilistic defect and the metric covering defect are parallel constructions:

$$H(X \mid Y) \quad \leftrightarrow \quad H_{\varepsilon/L}(X, d_X) - H_\varepsilon(\mathcal{O}(X), d_Y).$$

Both measure distinguishability lost under observation, one probabilistically and one metrically.

9 Observation design and bottlenecks

A mathematically controlled observation design problem has the form

$$\max_K I(Q;Y) \qquad \text{subject to} \qquad I(X;Y) \le R,$$

where $Q = f(X)$ is the target invariant and R is an information budget. This is the information-bottleneck form of a Geometry of Observation question: retain information relevant to the declared invariant while compressing irrelevant hidden detail.

A distortion version is

$$\min_K \mathbb{E}[d_Q(Q, \widehat{Q}(Y))] \qquad \text{subject to channel or coding constraints.}$$

This is not developed here, but it is the natural bridge to coding theory and rate-distortion theory.

10 Relation to other observation defects

Layer	Defect	Meaning
Information theory	$H(X \mid Y)$	hidden uncertainty inside observed fibers
Metric entropy	$H_{\varepsilon/L}(X) - H_\varepsilon(\mathcal{O}(X))$	lost covering distinguishability
Matroid theory	$r_{\text{hidden}} - r_{\text{observed}}$	lost independent structure
CSP / graph homomorphism	$F_{\mathcal{O}}$	unavoidable violated constraints after quotient
Tensorial geometry	$\ker(d\mathcal{O})$	vertical directions invisible to observation

11 Claim audit

Allowed statement	Not allowed without additional proof
Observation channels have conditional entropy and mutual information.	Information theory derives spacetime physics.
Data processing prevents post-processing from increasing information about a hidden source.	More data processing creates new hidden-state information.
Recoverable invariants are characterized by $H(f(X) \mid Y) = 0$.	Full state recovery follows from invariant recovery.
KL distinguishability contracts under stochastic observation.	Observed distinguishability is always equal to hidden distinguishability.
Hidden compact fibers have entropy cost after discretization.	A hidden fiber is automatically a physical extra dimension.

12 Conclusion

The information-theoretic layer gives a precise version of a central observation-geometry principle:

$$\boxed{\text{observation loses not only coordinates, but distinguishability.}}$$

For deterministic observations, the loss is fiber entropy $H(X \mid Y)$. For stochastic channels, mutual information and KL contraction measure retained and lost distinguishability. For declared invariants $Q = f(X)$, recoverability is exactly the condition $H(Q \mid Y) = 0$. This note therefore provides the probabilistic counterpart of the metric, matroidal, and constraint-theoretic defects developed in the surrounding Geometry of Observation program.

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